

HP Chromebook 14 G7

IT Intern Repair Guide

Ottumwa High School | IT Department

Version 2.0 | 2025-2026 School Year | By: James Ockenfels

Key Contacts

Supervisor: Jane Doe

IT Lead: John Doe

When in doubt, ask. Escalate all software, OS, enrollment, and unfixable hardware issues to John Doe or Jane Doe.

Quick Reference, Find Your Problem

Use this table to jump to the right section. Each section starts on its own page.

Problem / Symptom	Section to Go To
Cracked or broken screen	Issue 1, Screen Replacement
Keys broken or not responding	Issue 2, Keyboard Replacement
Not charging / no LED when plugged in	Issue 3, Charging Diagnosis
Charges but battery dies too fast	Issue 3, Battery Replacement
Top shell cracked or Wi-Fi antennas bad	Issue 4, Top Shell Replacement
Bottom shell cracked or damaged	Issue 5, Bottom Shell Replacement
Wi-Fi not working / slow network	Issue 6, Wi-Fi Chip Replacement
The device is slow or has a minor issue 5	Issue 7, Power Wash
Touchpad not working	Issue 8, Touchpad Replacement
Camera not working	Issue 9, Camera Replacement
Screen is black / no image	Issue 10, Display Cable Swap
Software problem / slow device	ESCALATE
ChromeOS missing or damaged	ESCALATE
Enrollment issues	ESCALATE
Device cannot be repaired	Part it, verify with Jane or John first

Rules You Must Know Before Starting Any Repair

- **Disconnect the battery before every repair unless stated otherwise. Do not reconnect it until the repair is fully complete.**
- **Working on the Wi-Fi chip with the battery connected will fry the chip.**
- **Do not touch or disturb the camera while it is in use, it will get fried.**
- **Do NOT Powerwash IHCC student devices. They have an authenticator that will break.**
- The keyboard and display cable can be removed while the device is on, but a mouse must be plugged in before startup to be recognized.
- Any device that cannot be repaired must be parted out, but verify first with Jane Doe or John Doe.
- Always update the ticket with proper ticket etiquette when working on a device that has one.
- The asset ID is on the sticker on the bottom of the device, or in the top-right corner of the login screen.

Tools

We use the iFixit Pro Repair Kit for all repairs. Make sure the kit is available before you start. You will mostly need:

- Jimmy / spudger / pry tool, for opening seams, cutting adhesive, and prying bezels
- Tweezers or forceps, for small cables and adhesive tabs
- Double-sided tape, thin strips, for screen replacement
- Parts cart: replacement screens, keyboards, batteries, Wi-Fi chips, shells, and cables are kept here

NOTE: ESD (static electricity) is worth being aware of. Avoid working on carpet if you can, and try not to touch circuit board traces or connector pins directly. We don't use wrist straps routinely but being mindful helps.

Parts Reference

These diagrams and component lists are sourced from the official HP Maintenance and Service Guide for the Chromebook 14 G7. They are included here so you can quickly identify parts by name and location. For more detailed part information, including additional diagrams, removal sequences, and specifications, refer to the full HP service manual at the link shown below.

Source: HP Official Service Manual

Diagrams for reference. All parts are kept on the parts cart, you do not need to look up part numbers for day-to-day repairs.

For more detailed part information, refer to the full HP Maintenance and Service Guide at:
https://kaas.hpcloud.hp.com/pdf-public/pdf_12616543_en-US-1.pdf

Figure 1: Major Components (Exploded View)

This diagram shows the main components of the Chromebook base assembly. Use it to identify part names and their locations inside the device.

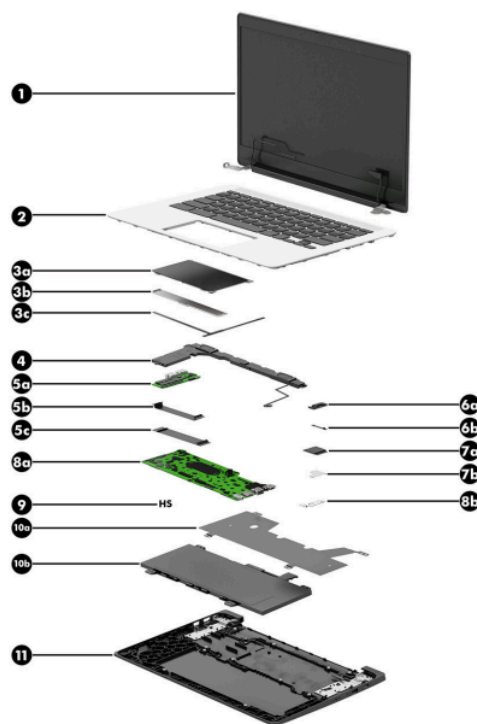


Table 1, Computer Major Components

1. Display assembly
2a. Top cover / keyboard without backlight (jet black finish)
3a. Touchpad

3b. Touchpad bracket
3c. Touchpad cable
4. Speaker (includes left and right speakers and cables)
5a. Connector board (includes HDMI port and USB port)
5b. Connector board low-power cable
5c. Connector board high-power cable
6a. Sensor board
6b. Sensor board cable
7a. Wi-Fi module (Intel Wi-Fi 6 AX201 with Bluetooth)
7b. WLAN module bracket
8. System board
9. Heat sink
10a. Battery shield
10b. Battery

Figure 2: Display Assembly Components

This diagram shows the parts that make up the display assembly (the lid). Use it to identify components during screen, camera, cable, and hinge repairs.

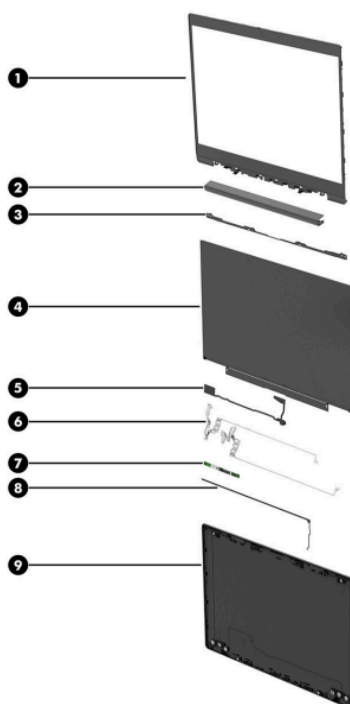


Table 2, Display Components

1. Display bezel
2. Display hinge cover
3. Display bracket
4. Display panel assembly
14.0" FHD touchscreen
14.0" FHD non-touchscreen
14.0" HD touchscreen
14.0" HD non-touchscreen
5. Display panel cable, for touchscreen models
Display panel cable, for non-touchscreen models
6. Display hinges (left and right)
7. Webcam / microphone module
8. Webcam / microphone module cable
9. Display back cover (includes wireless antenna)

Source: HP Official Service Manual

For complete part numbers, additional diagrams (cable routing, screw locations), and full disassembly sequences, see the HP Maintenance and Service Guide:

https://kaas.hpcloud.hp.com/pdf-public/pdf_12616543_en-US-1.pdf

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Issue 1, Screen Replacement

Do this when the display panel is physically cracked, has dead lines or pixels, or is confirmed non-functional. If the screen is simply black, check Issue 10 (Display Cable) first.

WARNING: A black screen alone does not mean the panel is bad. Test with an external monitor before replacing the screen.

1. Power off and unplug the device.
2. Open the lid. Use a pry tool to remove the hinge cover (the small plastic strip at the bottom of the display assembly).
3. Pry the bezel off, starting at a corner. Work slowly around all four edges, there are plastic clips. Go carefully to avoid cracking the bezel.
4. Look at the lower corners of the screen for small black pull-to-release adhesive tabs. If they are there, use tweezers to slowly roll them outward to release the screen.
5. If the tabs are missing, the screen has already been replaced before. Use a jimmy tool to cut along the edges of the replacement adhesive that was applied last time. Note: the original factory black adhesive cannot be cut, only replacement double-sided tape can be cut.
6. Once the screen is loose, carefully lay it face-down onto the keyboard. Do not yank, the display cable is still attached at the bottom.
7. Peel back the tape covering the display cable connector, then disconnect the cable by unlatching it and pulling straight back.
8. Take the new screen from the cart. Plug the display cable in and confirm it seats fully then latch it.
9. Before securing the screen: power on and verify the new screen works. Check for dead pixels, proper colors, and touch response if applicable. Power off before continuing.
10. Apply thin strips of double-sided tape to the metal frame around the outside edge of the screen. Do NOT put tape on the white light diffuser, only on the metal frame.
11. Route the display cable carefully at the bottom through its small channels so it is not pinched or kinked.
12. Peel the protective plastic off the new screen, then press it gently into place.
13. Snap the bezel back on around all edges, then press the hinge cover back on.

WARNING: Do not put tape on the white light diffuser area, it will cause bright spots on the display.

Issue 2, Keyboard Replacement

Do this when keys are physically broken, missing, or not registering. If the entire keyboard is dead, try reseating the cable first before doing a full swap.

1. Power off and unplug the device.
2. Flip the device upside down and remove all 8 bottom screws. The 2 screws at the back edge are slightly longer than the other 6, keeping them separate. Mixing them up is a common mistake that strips the keyboard tray and damages the keyboard.
3. Open the lid and lay the device flat on the desk.
4. Use a pry tool to work around the seam between the keyboard and the base to pop the keyboard assembly up. Work around the edges until it lifts.
5. Lift the front edge of the keyboard just enough to access the ribbon cables underneath, do not yank it fully off.
6. Peel back the small plastic film covering the cable connectors. Each connector (keyboard cable and touchpad cable) has a small latch, lift the latch with a fingernail or spudger, then slide the cable out straight.
7. Remove the old keyboard assembly.
8. Take the replacement keyboard from the cart. Slide each ribbon cable fully into its connector and press the latch back down to lock it.
9. Before screwing anything in: power on and test every key and the touchpad. Fix any connection issues now while everything is open.
10. For testing, you can use the built-in keyboard tester that can be found on the diagnostic app built into the Chromebooks.
11. Once confirmed working, fold the keyboard down into place, flip the device over, and replace all 8 screws, the 2 longer screws go in the back edge holes only.

WARNING: The 2 longer screws go in the rear holes ONLY. Installing a long screw in the wrong hole will thread into the plastic keyboard tray, damage it, and leave a bump under the plastic.

Issue 3, Charging Diagnosis and Battery Replacement

The LED on the right side of the device glows amber when charging and white when full. If neither light appears, follow these steps.

Step A, Rule Out the Charger

Test with a different known-good USB-C charger first. This takes 30 seconds and is the most common fix.

Step B, Disconnect the Battery and Test

1. Follow Issue 2 to remove the keyboard and access the internals.
2. Locate and disconnect the battery cable from the system board.
3. Plug the charger back in and watch the LED indicator.

LED Result	What It Means
LED does NOT light up	USB-C port or motherboard is damaged. Not repairable in-house. Part the device after verifying with Jane or John.
LED lights up	Power is reaching the board. Try booting. If it boots, the battery is bad, replace it (see Step C below).

Step C, Replace the Battery

If the device boots with the battery disconnected, the battery is bad and needs replacing.

1. With the keyboard already removed, locate the large flat battery in the center of the base.
2. Remove the battery shield screws and lift the shield off. Set it aside, you will reuse it.
3. Remove the battery.
4. Install the replacement battery from the cart, reconnect the cable, and reuse the same shield and screws.
5. Reinstall the keyboard, plug in the charger, and confirm the LED behavior is correct.

TIP: A battery that drains much faster than normal is also a sign it needs replacing. Check battery health in the diagnostics app.

Issue 4, Top Shell Replacement

Do this when the top shell (lid) is physically damaged, or when Wi-Fi antenna problems persist after a Wi-Fi chip swap. The antennas are built into the top shell.

WARNING: Disconnect the battery before this repair and do not reconnect it until everything is fully reassembled.

Disassembly

1. Remove the keyboard (Issue 2).
2. Remove the hinge cover, then the bezel, then the screen (Issue 1 steps 2 through 7).
3. Remove the small metal bar at the top of the shell. Note how the metal hinge tabs fit under it at the screw points, you will need to replicate this on reassembly.
4. Carefully detach the camera from its mount, noting that the camera cable runs underneath it. Disconnect the camera cable.
5. Remove the speaker (3 screws).
6. Remove the Wi-Fi chip bracket and disconnect both Wi-Fi antenna cables and the display cable, and the camera cable from the motherboard side.
7. Work the camera cable, both Wi-Fi antenna cables, and the display cable back through the hinges toward the top shell.
8. Remove the 4 screws holding each hinge to the top shell (4 per side, 8 total).
9. Shimmy the top shell off, feeding the cables through as you go. Set aside the camera, camera cable, display cable, metal bar, and screen, you will reuse all of these.

Reassembly

10. Take a new top shell from the cart. The new shell comes with Wi-Fi antennas already installed.
11. Install the camera and camera cable into the new shell.
12. Line the top shell up with the hinges. Carefully route the camera cable, both Wi-Fi antenna cables, and the display cable through the small gaps in the hinges.
13. Once cables are properly routed, screw in the 4 hinge screws on each side.
14. Install the screen (Issue 1 steps 8 onward).
15. Install the metal bar at the top, make sure the small metal hinge tabs sit underneath the bar at the screw points before tightening.
16. Install the bezel, then the hinge cover.
17. On the motherboard side, plug in all 4 cables (camera, both Wi-Fi antennas, display) and make sure they are routed correctly and not pinched.
18. Install the Wi-Fi chip bracket, then the speaker, then close up and reinstall the keyboard.

WARNING: Incorrect cable routing will cause cables to tear when opening and closing the device, and can prevent the hinge cover from fitting properly.

Issue 5, Bottom Shell Replacement

Do this when the bottom shell is cracked or physically damaged. Note: the serial number etched into the bottom of the device will be different after this repair.

WARNING: Disconnect the battery before this repair and do not reconnect until fully reassembled.

Disassembly

1. Remove the keyboard (Issue 2).
2. Remove the speaker (3 screws) and the Wi-Fi chip bracket.
3. Disconnect the camera cable, display cable, and both Wi-Fi antenna cables from the motherboard.
4. Remove the screws holding the motherboard, USB-C bracket, daughterboard, Wi-Fi chip, and battery.
5. Disconnect and remove the battery.
6. Carefully lift out the motherboard and daughterboard. You may want to disconnect the cable linking them before removing.
7. Unscrew the 2 screws on each hinge (both sides).
8. Carefully close the device. Then slide the top shell and bottom shell in opposite directions to separate them.

Reassembly

9. Take a new bottom shell from the cart. Use the same sliding technique in reverse to join the top shell to the new bottom shell.
10. Carefully open the device slowly, pressing each hinge down manually as you go, the hinges are not yet secured to the bottom shell.
11. Screw the hinges into place on both sides.
12. Reinstall the motherboard and daughterboard, making sure they are connected to each other.
13. Install the battery and reconnect the battery cable.
14. Install the Wi-Fi chip.
15. Reconnect all 4 cables (camera, display, both Wi-Fi antennas) and make sure they are routed correctly.
16. Install the Wi-Fi chip bracket and the speaker.
17. Reinstall the keyboard.

Issue 6, Wi-Fi Chip Replacement

Do this for devices with persistent network issues that were not fixed by a Powerwash, or that cannot be Powerwashed because they belong to an IHCC student.

WARNING: Working on the Wi-Fi chip with the battery plugged in WILL fry the chip. Always disconnect the battery first.

NOTE: Do NOT Powerwash IHCC student devices. Their devices have an authenticator that will permanently break if Powerwashed. For IHCC students with Wi-Fi issues, go straight to this chip replacement without Powerwashing.

1. Remove the keyboard (Issue 2) and disconnect the battery.
2. Remove the Wi-Fi chip bracket (small cover held by 1 screw).
3. Disconnect the two small antenna cables from the chip. Note that they are numbered and will need to go back into the same numbered ports.
4. Remove the chip screw and lift the old chip out.
5. Install a replacement chip from the cart. Reconnect the antenna cables in the correct positions.
6. If the antenna cables won't seat properly on the new chip, try another chip from the cart. If no chip works with the antennas, the top shell likely needs replacing (Issue 4).
7. Once the chip is in and cables are connected, reinstall the bracket. Reconnect the battery, close up, and test the internet connection.
8. If the issue persists after both a chip swap and top shell replacement, the device needs a new unit. Part it out after verifying with Jane or John.

Issue 7, Powerwash (Factory Reset)

A Powerwash clears all local data and resets ChromeOS to a clean state. Try this first for slowness, app crashes, or minor software issues. It does NOT delete anything in Google Drive.

WARNING: Do NOT Powerwash IHCC student devices. Their authenticator will break and they will lose access to their accounts.

Method 1, Developer Mode Trick (Our School's Method)

This is the method set up by the school. It triggers a Powerwash rather than actually enabling developer mode.

- Press Esc + Refresh + Power at the same time to enter recovery mode.
- Press Ctrl + D when prompted.
- Hit Enter a few times when asked. The school has it configured to Powerwash instead of enabling developer mode.

Method 2, Keyboard Shortcut

From the login screen (not while signed in), press Shift + Alt + Ctrl + R. A Powerwash prompt will appear. Follow the on-screen steps.

NOTE: The Settings menu method does not work on some school-managed devices. Use Method 1 or Method 2 above.

After a Powerwash completes, the device will ask for a Google account login. Make sure the student signs in with their school account so it re-enrolls on the school network correctly.

If a Powerwash does not fix the issue, or the device cannot be Powerwashed, escalate to John Doe or Jane Doe. Do not attempt OS recovery or enrollment changes yourself.

Issue 8, Touchpad Replacement

Do this when the touchpad is physically damaged or confirmed not working after reseating the cable.

1. Remove the keyboard (Issue 2).
 2. Flip the keyboard assembly face-down.
 3. Remove the 4 screws holding the touchpad bracket, then the 3 screws connecting the touchpad directly to the keyboard tray.
 4. Disconnect the touchpad cable and remove the old touchpad.
 5. Get a replacement touchpad from the cart. Install in reverse order: connect the cable, set the touchpad in place, screw in the 3 direct screws, then attach the bracket with its 4 screws.
 6. Reinstall the keyboard and test the touchpad before finishing.
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Issue 9, Camera Replacement

Do this when the camera is not working. Follow these steps in order, start simple before doing a full cable replacement.

WARNING: Do not disturb or disconnect the camera while it is actively in use. This will fry it.

Step A, Check the Cable First

On the motherboard side, verify the camera cable port is plugged in securely and doesn't look damaged. If it looks fine and the camera still doesn't work, continue.

Step B, Replace the Camera Module

1. Remove the hinge cover and bezel (no need to remove the whole screen).
2. Carefully remove the camera module, note that the cable runs underneath it.
3. Install a replacement camera from the cart and test. If the camera now works, reinstall the bezel and hinge cover and you're done.
4. If the camera still doesn't work, continue to Step C.

Step C, Replace the Camera Cable

Before removing anything, test: plug the suspect cable into the motherboard and plug a working camera into the other end. If the camera works, the cable is fine and there is another issue. If it doesn't work, the cable is bad.

1. Remove the left hinge screws, the screen, the speaker, and the metal bar at the top of the top shell.
2. Remove the old cable, noting exactly how it is routed.
3. Install the new cable following the same path and routing.
4. Test that the camera works before reassembling.
5. If it works, reinstall the screen, speaker, metal bar, bezel, and hinge cover.
6. If the camera still does not work after a new module and new cable, escalate to John Doe or Jane Doe.

Issue 10, Display Cable Swap

Do this when the screen won't turn on at all, or turns on but has no backlight. Try the cable swap to rule it out before assuming the motherboard is bad.

Step A, Quick Test

1. Remove the keyboard (Issue 2). Unplug the existing display cable from the motherboard.
2. Plug in a test cable from the cart and connect a test screen to it. If the screen lights up and works, the original cable is bad, continue to Step B. If the test screen also doesn't work, the motherboard is likely damaged. Escalate to John Doe or Jane Doe.

Step B, Full Cable Replacement

3. Remove the hinge cover, bezel, and screen (Issue 1 steps 2 through 7).
 4. Remove the 4 screws from the right hinge on the top shell.
 5. Pull the old display cable out, noting its routing path.
 6. Route the new cable the same way and reconnect everything.
 7. Test the screen and display cable together before reassembling.
 8. If it works, screw the 4 hinge screws back in, reinstall the screen, bezel, hinge cover, and keyboard.
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When to Escalate, and When to Part a Device

Always Escalate to John Doe or Jane Doe

- Any software issue, OS problem, or ChromeOS error screen
- Enrollment issues or anything with Google Admin
- Damaged motherboard or USB-C port
- Any repair you are unsure about, ask before you make it worse

Parting a Device

If a device truly cannot be repaired, it should be parted out so its components can be used on future repairs.

- Always verify with Jane Doe or John Doe before parting a device.
- Useful donor parts: screens, keyboards, batteries, Wi-Fi chips, cameras, cables, shells.
- Label the parted components clearly and store them on the parts cart.

General Tips

- Always test a repair before fully reassembling. It is much easier to fix a connection while everything is open.
- Take a photo before disconnecting cables or removing components so you have a reference.
- If a screw is stripped, don't force it, ask for help.
- Update the ticket using proper ticket etiquette any time you work on a device that has one.
- The asset ID is on the sticker on the bottom of the device, or shown in the top-right corner of the login screen.
- Use a magnetic bit to simulate opening and closing the lid. Place the magnet over the sensor on the left side of the logic board for a few seconds, then remove it. This "tricks" the system into waking up or power cycling, which is especially useful for testing a device without the bezel or keyboard attached.

Good luck, and welcome to the IT team.

If you read this far your a nerd